

GoodWin's Quest!

Rewiring Recovery, One
Signal at a Time

♡ Team Chattar Pattar



What is the PROBLEM?

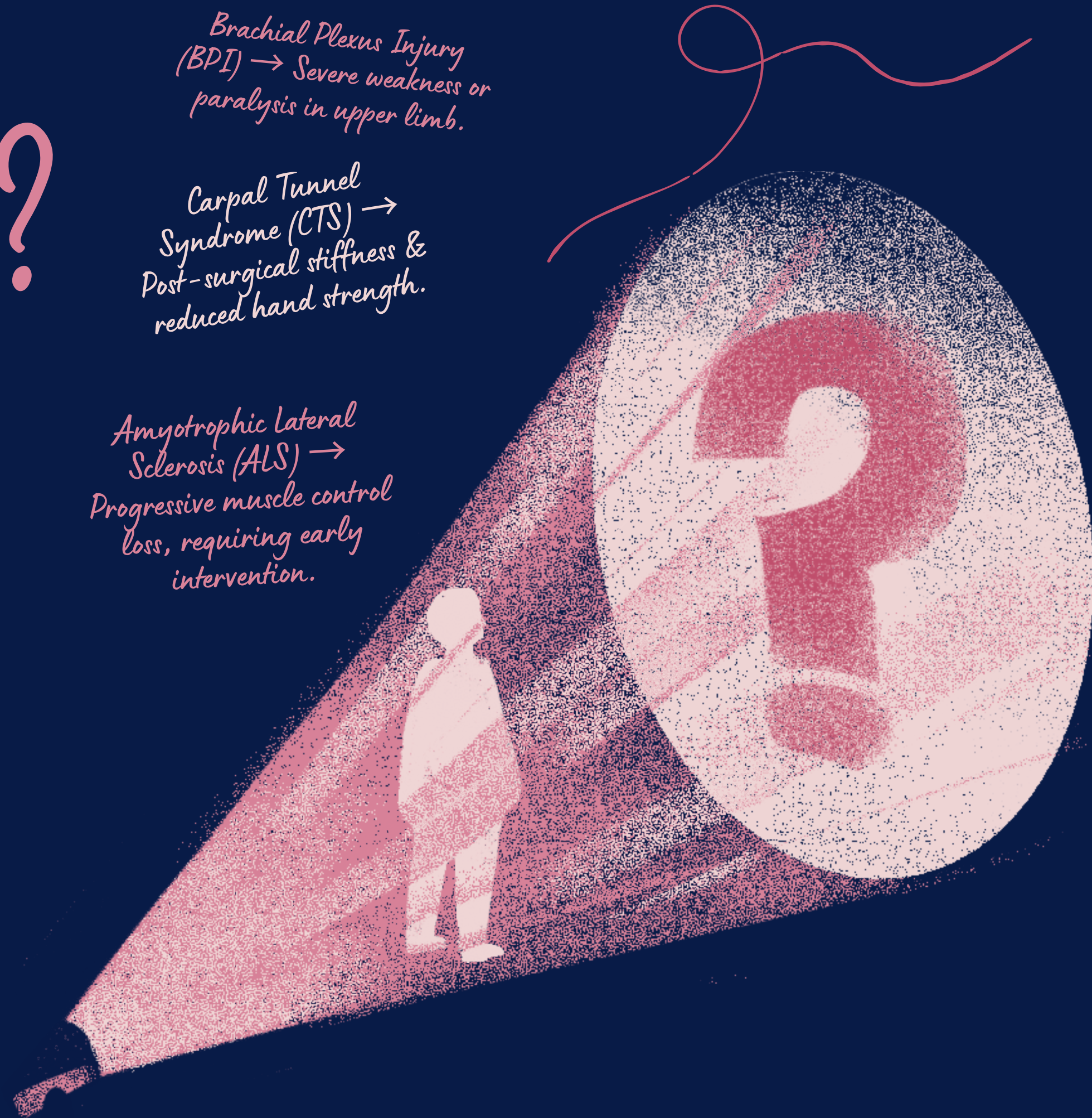
Millions worldwide suffer from hand mobility impairments due to neurological or musculoskeletal disorders.

Traditional therapy: repetitive, unengaging, lacks objective measurement.

Brachial Plexus Injury (BPI) → Severe weakness or paralysis in upper limb.

Carpal Tunnel Syndrome (CTS) → Post-surgical stiffness & reduced hand strength.

Amyotrophic Lateral Sclerosis (ALS) → Progressive muscle control loss, requiring early intervention.



Our Solution

*Immersive VR-based
rehabilitation platform for
interactive hand exercises.*

*Integration with EEG & EMG
for real-time monitoring of
brain-muscle activity.*

*Web-based system with
live video sessions &
automatic recording.*



*Adaptive therapy → Exercises
adjust to patient progress.*

*Provides objective,
measurable recovery data
for patients and therapists.*



Uniqueness and Advantages

Goes beyond VR rehab → integrates neuro-muscular biosignals

Enables objective tracking of recovery, not just movement observation.

Adaptive AI-driven difficulty levels for personalized therapy

Remote monitoring allows rehabilitation beyond hospital walls.

Data-driven insights → useful for both patients and clinicians.

Key Features of the Website

Live Rehabilitation Sessions

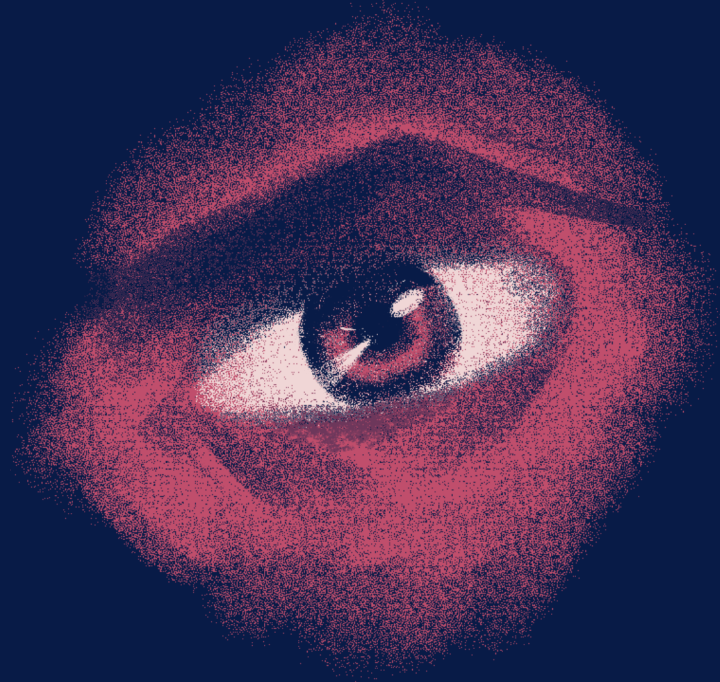
- *Video streaming of exercises with therapist guidance.*
- *Auto-recording & cloud storage of sessions.*

EEG & EMG Analytics Dashboard

- *Real-time brain & muscle activity graphs.*
- *Fatigue detection, coordination analysis.*

Patient & Therapist Dashboards

- *Progress tracking with visual comparisons.*
- *Secure clinician access to session data & reports.*



Competitive Differentiation

Current VR rehab systems
→ limited to motion
tracking & gamification.

Our system → captures
brain-muscle coordination
(EEG + EMG)



Provides deeper clinical
insights into recovery
progress

Combines video-based
therapy, biosignal
analytics, and progress
dashboards in one
platform.

Clinical & Social Impact

For Patients
Engaging, motivating, personalized therapy.

For Clinicians
Objective, data-backed progress tracking.

For Society
Reduced recovery time, improved quality of life.



Business Model Canvas

8. KEY PARTNERS [1]

- Hospitals & rehabilitation centers
- Neurologists, physiotherapists, occupational therapists
- Medical device manufacturers (EEG/EMG sensors)
- VR hardware providers (Oculus, HTC Vive, etc.)
- Research institutions & universities
- Insurance companies & healthcare payers
- Cloud/AI technology providers

6. KEY ACTIVITIES [2]

- Development & maintenance of VR rehab platform
- Integration of EEG, EMG & biosignal analytics
- Conducting clinical trials & validations
- Training therapists & onboarding patients
- Data security & compliance (HIPAA, GDPR)
- Customer support & technical assistance

7. KEY RESOURCES [6]

- VR software & gamified rehab modules
- EEG & EMG hardware and signal processing algorithms
- Cloud infrastructure & data analytics dashboard
- Clinical expertise from neurologists/rehab specialists
- R&D team for AI-driven personalization
- Patents / IP for biosignal integration

2. VALUE PROPOSITIONS [3]

- For Patients → Engaging, adaptive, and measurable rehabilitation experience
- For Clinicians → Objective biosignal-driven insights into recovery progress
- For Hospitals → Scalable tele-rehabilitation solution reducing in-clinic load
- For Insurers → Evidence-based outcomes to justify reimbursements
- Uniqueness: Combines VR + EEG/EMG + analytics + video records in one integrated solution

5. CUSTOMER RELATIONSHIPS [4]

- Personalized therapy plans & dashboards
- Remote guidance & tele-consultations
- Automated progress reports for patients & clinicians
- Continuous engagement through gamified rehab modules
- 24/7 technical & medical support

4. CHANNELS [7]

- Direct partnerships with hospitals & rehab centers
- B2B licensing for clinics & physiotherapy practices
- Web platform & patient mobile app
- Conferences, medical exhibitions, research publications
- Online marketing to patient communities & caregivers

1. CUSTOMER SEGMENTS [5]

- Primary: Patients with CTS, BPI, ALS
- Secondary: Hospitals, rehab centers, clinics
- Tertiary: Insurance providers, researchers in neurorehab

9. COST STRUCTURE [8]

- R&D for VR software, signal processing, AI
- Hardware procurement (EEG/EMG sensors, VR headsets)
- Cloud hosting & data storage
- Clinical validation trials
- Marketing & awareness campaigns
- Regulatory certifications (FDA, CE, etc.)

3. REVENUE STREAM [9]

- Subscription model for patients (monthly/annual rehab plans)
- Licensing model for hospitals & clinics (per seat/device)
- Pay-per-session tele-rehabilitation packages
- Data-driven insights for research institutions (with consent)
- Insurance reimbursements (partnered models)
- Hardware bundles (VR headset + EEG/EMG kit + software license)